

Standing Waves and Superposition

1 Starting point: large rope

2 When Waves Meet

Guiding Question

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Yesterday we learned that sound consists of travelling pressure and density waves. When you pluck a guitar string, you can hear a clear note for several seconds.

Discussion

- If sound waves simply travel away, why does the note last for several seconds?
- What must happen inside the instrument while we continue to hear the sound?
- Can a single travelling wave explain this observation?

Expected ideas

- The string (or the air inside the instrument) must continue vibrating.
- The vibration cannot simply leave the instrument immediately.
- There must be a wave pattern that remains inside the instrument while sound is continuously radiated into the surrounding air.

Conclusion Something must happen that keeps the vibration inside the instrument instead of letting it simply travel away. A travelling wave transports the disturbance through the medium. It does not remain in one place. Therefore, a different type of wave is needed to explain how a musical instrument can produce a long, steady note.

Activity 1: A Travelling Wave

A long rope is stretched along the room, a single pulse is generated.

Discussion

- What happens to the pulse?
- Does it stay in one place?
- What happens after it reaches the end of the rope?

Expected answers

- The pulse travels along the rope.
- It does not stay in one place.
- Each point of the rope moves only for a short time before returning to rest.

- When the pulse reaches the end, it disappears (or is reflected, depending on the setup).
- The disturbance moves through the rope, not the rope itself.

Now, a harmonic wave is generated. **Discussion**

- What is different from a single pulse?
- Does the wave remain on the rope?
- Could this kind of wave produce a steady musical note? Why or why not?

Expected answers

- The pulse travels along the rope.
- It does not stay in one place.
- Each point of the rope moves only for a short time before returning to rest.
- When the pulse reaches the end, it disappears (or is reflected, depending on the setup).
- The disturbance moves through the rope, not the rope itself.

Students will notice that the wave simply travels along the rope and eventually disappears at the end.

Activity 2: Reflection

Generate a single pulse with reflection at the end

Discussion

- What happens when the pulse reaches the end?
- Does it disappear completely?
- Where does the reflected pulse travel?

Expected answers

- The pulse is reflected.
- It travels back along the rope.
- For a short time, there are two pulses travelling in opposite directions.

Sketch on the board:

→ ← Question:

- How many waves are now travelling on the rope?

Expected answer:

- Two waves travelling in opposite directions.

Activity 3: Superposition in the Ripple Tank

Start with a **single wave source**, just for a few seconds.

observation:

- transversal wave
- spherical direction of travel
- in the ripple tank: water can spread in two dimensions; can move forward, backward and left right
- rope behaves like a line out of the ripple tank
- only difference: wave can travel in two directions in water but only in one direction in the rope

If possible: plain waves which are reflected by a straight barrier otherwise switch on two point sources.

Discussion

- What do you notice?
- Are all parts of the water moving equally?
- Why are some regions almost calm?
- Why are other regions moving much more strongly?

Expected answers

- Two waves can exist at the same place.
- Sometimes the waves reinforce each other.
- Sometimes they cancel each other.
- The total displacement is the sum of both waves.

Introduce the principle of **superposition**.

Whenever two waves meet, their displacements simply add together.

3 back to rope

: We can also see this phenomenon with ropes $\Rightarrow$ standing waves:

Discussion

- What do you notice?
- Which points never move?
- Which points move the most?
- (Can you explain these observations using the idea of superposition?)

Expected answers

- Some points always cancel because the two waves have opposite displacements.

- These points are called **nodes**.
- At other points the two waves always reinforce each other.
- These points are called **antinodes**.

Conclusion

A standing wave is not a new type of wave.

It is formed when two identical harmonic travelling waves move in opposite directions and continuously superpose.

The incoming wave and the reflected wave together produce the standing wave pattern.

4 Do standing waves occur for all frequencies/wavelengths??

Form groups, each group gets a rope. Tasks:

- Find as many stable wave patterns as possible.
- Draw them.
- Count the antinodes.
- optional: Measure the wavelength.
- optional: Which frequencies work?
- optional: Which frequencies do not work?
- Can you find a rule?

beginitemize

Only certain wavelengths produce stable standing waves.

The wavelength must fit exactly onto the string.

As the number of antinodes increases, the wavelength becomes shorter.

Rule

A standing wave can only exist if an integer number of half wavelengths fits exactly into the length of the string. (je nach festem/losem Ende anpassen)

5 Seilwellenmaschine

6 guitar string

- The fundamental mode has the largest wavelength.
- Higher modes have shorter wavelengths.
- Since the wave speed is constant, $c = f\lambda$, a shorter wavelength means a higher frequency.
- short strings produce higher frequencies

Conclusion

Different standing wave patterns correspond to different wavelengths. Because the wave speed remains constant, each wavelength corresponds to one particular frequency.

These frequencies are the musical notes that we hear.

7 second day

Guiding Question

We have discovered that a stretched string can only support certain standing waves.

Discussion

- Do you think standing waves only exist on strings?
- What other parts of a musical instrument could support standing waves?

Expected answers

- Air columns.
- The body of an instrument.
- Any medium in which waves can travel.

not all musical instruments use strings, some musical instruments also use vibrating air columns

- string only allows certain standing waves
- both ends are fixed $\rightarrow$ no oscillation possible $\rightarrow$ nodes
- recorder: Does it have two fixed ends? - no
- depending on the instrument, we have different combinations of fixed/open ends - called boundary conditions
- two fixed ends: guitar
- two open end: transverse flute
- one open and one fixed end: pan flute

different instruments have different ends and support different wavelength/ frequencies

Group Investigation

Divide the class into three groups.

Each group investigates a different type of musical instrument.

Group 1: String (fixed – fixed)

Examples:

- Guitar
- Violin
- Piano

Tasks

- Draw the first three standing waves.
- Mark the nodes and antinodes.

- Which wavelengths fit onto the string?
- Which frequencies are possible?

Group 2: Open Pipe (open – open)

Examples:

- Flute
- Open organ pipe

Tasks

- Draw the first three standing waves.
- Mark the nodes and antinodes.
- Which wavelengths fit into the pipe?
- Which frequencies are possible?

Group 3: Closed Pipe (closed – open)

Examples:

- Bottle
- Clarinet (approximately)
- Closed organ pipe

Tasks

- Draw the first three standing waves.
- Mark the nodes and antinodes.
- Which wavelengths fit into the pipe?
- Which frequencies are possible?
- What is different from the open pipe?

Instrument	Ends	Smallest wave	Higher modes
Guitar string	Fixed – Fixed	Half a wavelength	1, 2, 3, 4, ...
Flute	Open – Open	Half a wavelength	1, 2, 3, 4, ...
Bottle / Clarinet	Closed – Open	Quarter wavelength	1, 3, 5, ...

The standing wave with the lowest possible frequency is called the **fundamental mode** (or **fundamental frequency**).

It determines the pitch of the musical note that we hear.

Standing waves with higher frequencies are called **harmonics** (or **overtones**). We will investigate their significance in the next days.

8 game - optional

each student is either a node or anti-node. - They have to build standing waves for the different harmonics and different boundary conditions.